job_7-Evaluation report

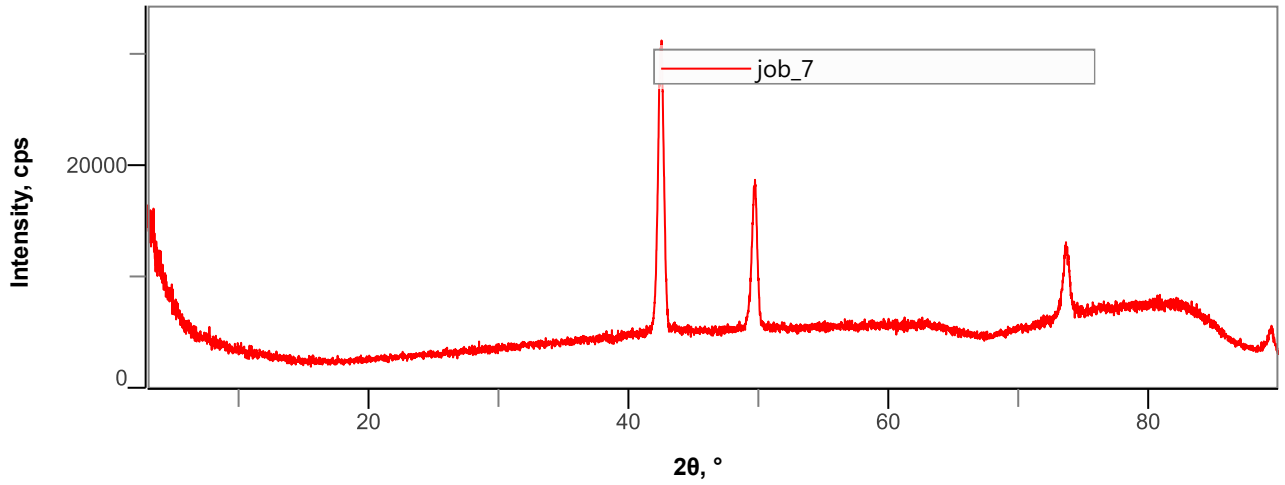
General information

Analysis date	2025-12-17 10:58:05	Measurement start time	2025-11-23 10:25:24
Analyst	Administrator	Operator	Administrator
Sample name	job_7	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_7.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

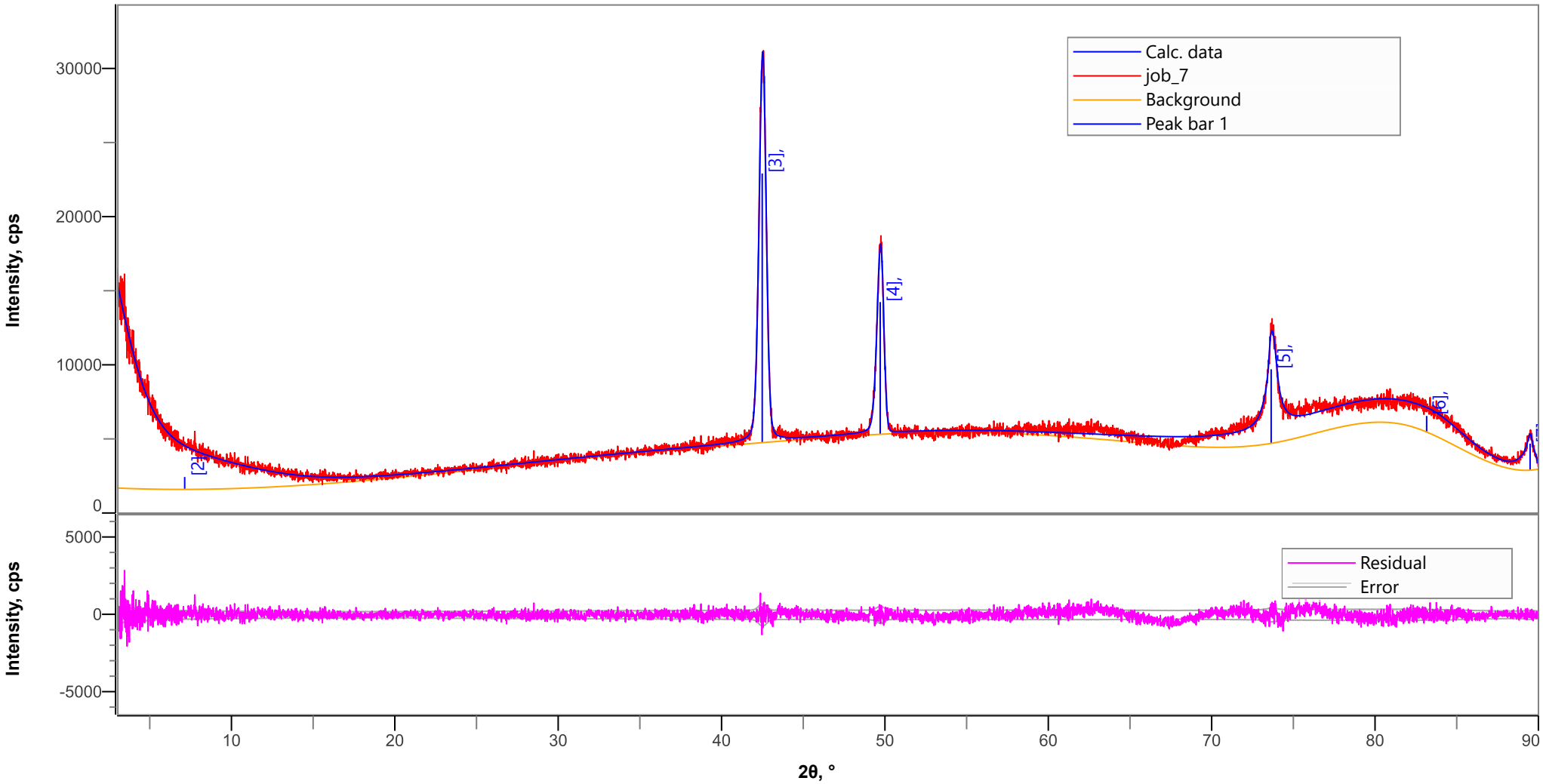
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

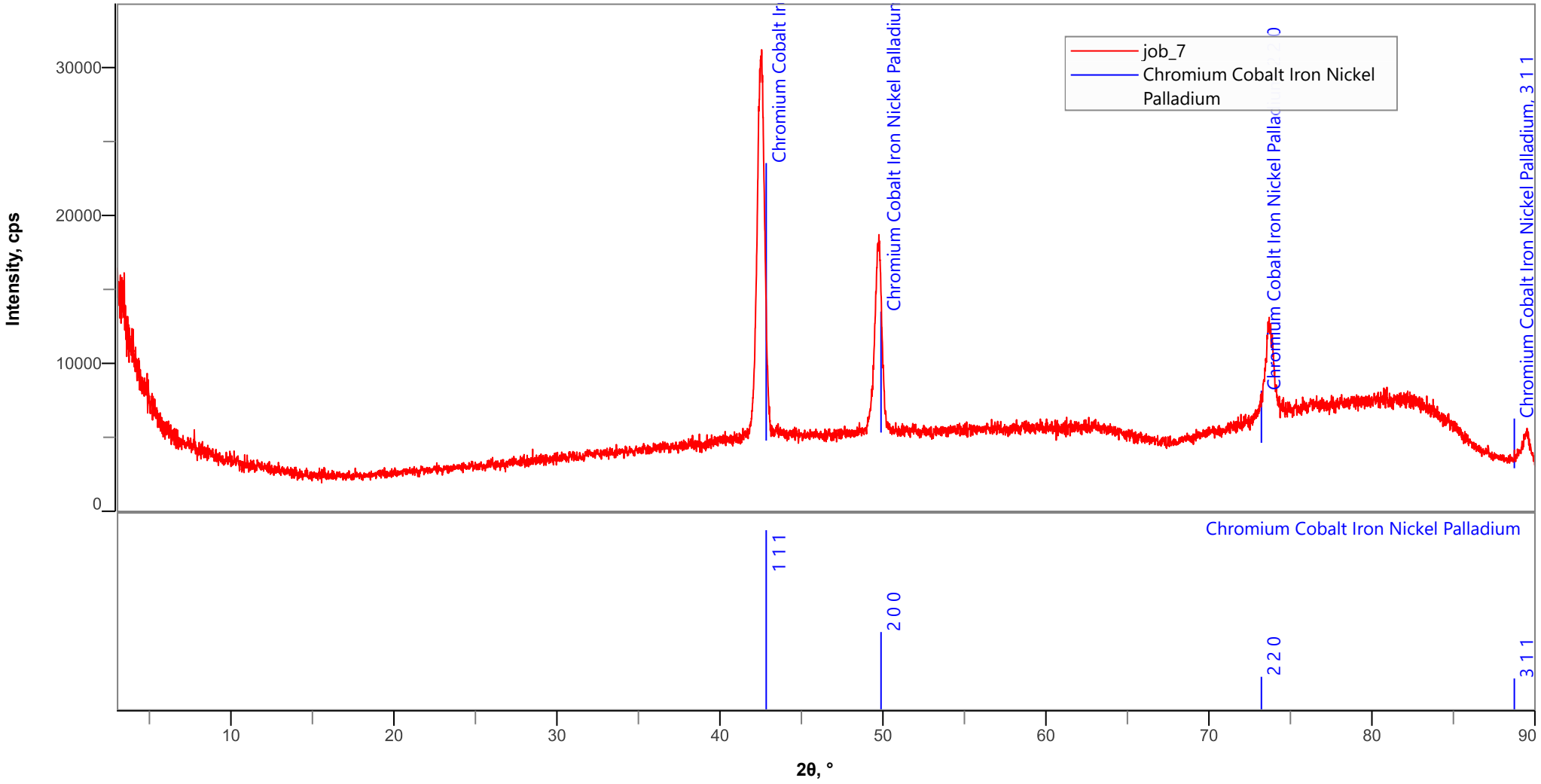
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Chromium Cobalt Iron Nickel Palladium	Cr0.20 Fe0.20 Co0.20 Ni0.20 Pd0.20	0.725	S/M:PDF-5+ 2026	225 : Fm-3m	04-018-7505

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, cps	FWHM, °	Int. I., cps°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay (ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	1.6 (10)	54 (34)	11746 (5200)	9.5 (18)	127636 (56503)	11 (10)	3 (278159)	0.0 (19)	0.8(3)	8.8(17)	Unknown			100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	7.1 (12)	12 (2)	835 (50)	11.8 (14)	10693 (3451)	13 (5)	1 (147)	0 (23)	0.0(3)	7.0(9)	Unknown			8.38	Split Pseudo-Voigt	-	-	-	-	-	
3	42.468 (3)	2.12685 (16)	18135 (424)	0.483 (3)	10265(54)	0.566 (16)	1.09(3)	0.31 (2)	0.27(2)	184.4 (10)	Chromium Cobalt Iron Nickel Palladium: 1 1 1	Cr0.20 Fe0.20 Co0.20 Ni0.20 Pd0.20	04-018-7505	8.04	Split Pseudo-Voigt	-	-	-	-	-	
4	49.685 (6)	1.8335 (2)	8909 (269)	0.455 (5)	4615 (48)	0.52 (2)	1.34(7)	0.28 (4)	0.08(5)	201(2)	Chromium Cobalt Iron Nickel Palladium: 2 0 0	Cr0.20 Fe0.20 Co0.20 Ni0.20 Pd0.20	04-018-7505	3.62	Split Pseudo-Voigt	-	-	-	-	-	
5	73.603 (14)	1.2859 (2)	4991 (181)	0.53 (3)	4621 (62)	0.93 (5)	0.92 (14)	1.27 (9)	1.15(10)	195(10)	Chromium Cobalt Iron Nickel Palladium: 2 2 0	Cr0.20 Fe0.20 Co0.20 Ni0.20 Pd0.20	04-018-7505	3.62	Split Pseudo-Voigt	-	-	-	-	-	
6	83.11 (5)	1.1612 (6)	1082 (51)	14.9 (2)	20060 (1172)	19 (2)	4.6 (4)	0.43 (12)	0.54(17)	7.45(11)	Unknown			15.72	Split Pseudo-Voigt	-	-	-	-	-	
7	89.44 (2)	1.0947 (2)	1789 (99)	0.49 (4)	1392 (89)	0.78 (9)	2.2 (5)	1.28 (13)	0.2(3)	241(22)	Unknown			1.09	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Chromium Cobalt Iron Nickel Palladium	3.65501	3.65501	3.65501	90.000	90.000	90.000

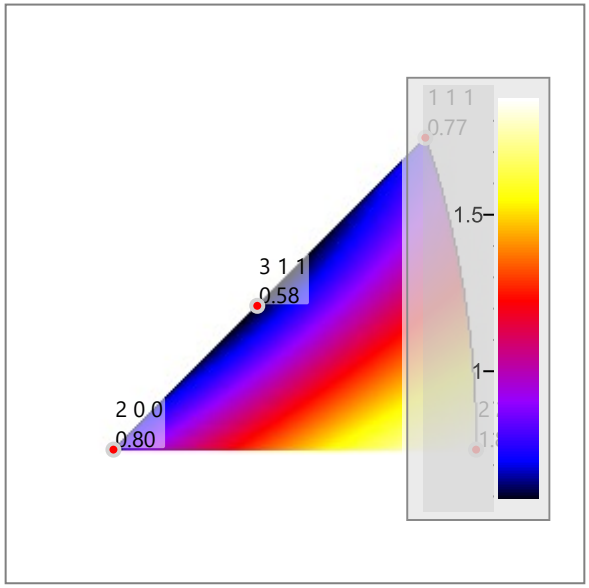
d-I List

Chromium Cobalt Iron Nickel Palladium

No.	2θ, °	d, Å	h k l	Norm. I.
1	42.81906	2.11022	1 1 1	100.00
2	49.85893	1.82750	2 0 0	45.53
3	73.18134	1.29224	2 2 0	23.77
4	88.69083	1.10203	3 1 1	26.57

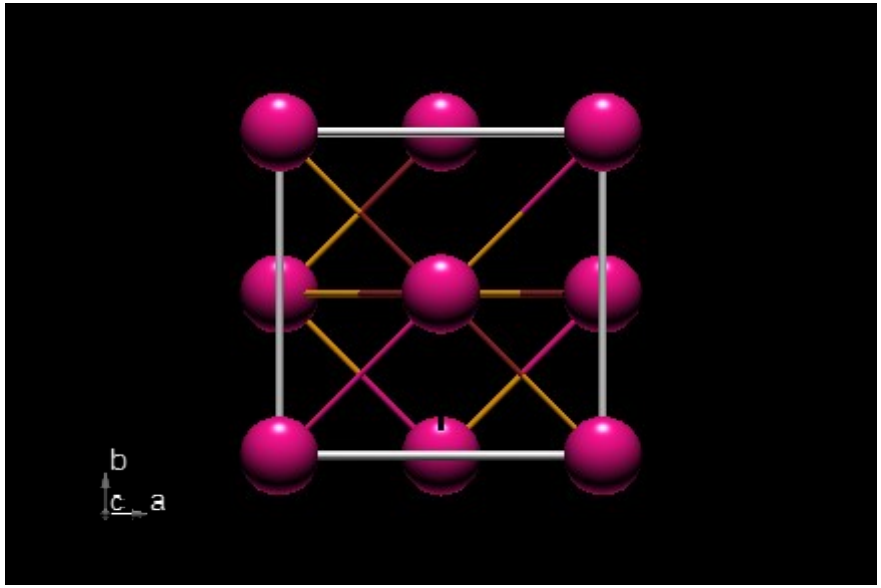
Simple Inverse Pole Figure

Chromium Cobalt Iron Nickel Palladium



Crystal Structure

Chromium Cobalt Iron Nickel Palladium



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Co1(Co)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000
Cr1(Cr)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000
Fe1(Fe)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000
Pd1(Pd)	0.000000	0.000000	0.000000	0.200	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Co1-Pd1(x,1/2+y,1/2+z)	2.58448
Co1-Pd1(1/2+x,y,-1/2+z)	2.58448
Co1-Pd1(-1/2+x,y,1/2+z)	2.58448
Co1-Pd1(1/2+x,y,1/2+z)	2.58448
Co1-Pd1(1/2+x,-1/2+y,z)	2.58448
Co1-Pd1(-1/2+x,1/2+y,z)	2.58448
Co1-Pd1(1/2+x,1/2+y,z)	2.58448
Co1-Pd1(x,1/2+y,-1/2+z)	2.58448
Co1-Pd1(x,-1/2+y,1/2+z)	2.58448
Co1-Ni1(1/2+x,y,-1/2+z)	2.58448
Co1-Ni1(-1/2+x,y,1/2+z)	2.58448
Co1-Ni1(x,1/2+y,1/2+z)	2.58448
Co1-Ni1(1/2+x,-1/2+y,z)	2.58448
Co1-Ni1(-1/2+x,1/2+y,z)	2.58448
Co1-Ni1(1/2+x,y,1/2+z)	2.58448
Co1-Ni1(1/2+x,1/2+y,z)	2.58448
Co1-Ni1(x,1/2+y,-1/2+z)	2.58448
Co1-Ni1(x,-1/2+y,1/2+z)	2.58448
Co1-Co1(x,1/2+y,1/2+z)	2.58448
Co1-Co1(x,-1/2+y,1/2+z)	2.58448
Co1-Co1(x,1/2+y,-1/2+z)	2.58448
Co1-Co1(1/2+x,1/2+y,z)	2.58448
Co1-Co1(-1/2+x,1/2+y,z)	2.58448
Co1-Co1(1/2+x,-1/2+y,z)	2.58448
Co1-Co1(-1/2+x,y,1/2+z)	2.58448
Co1-Co1(1/2+x,y,-1/2+z)	2.58448

Distance

Bond Def.	Distance, Å
Co1-Co1(1/2+x,y,1/2+z)	2.58448
Co1-Fe1(-1/2+x,y,1/2+z)	2.58448
Co1-Fe1(1/2+x,y,1/2+z)	2.58448
Co1-Fe1(1/2+x,-1/2+y,z)	2.58448
Co1-Fe1(-1/2+x,1/2+y,z)	2.58448
Co1-Fe1(1/2+x,1/2+y,z)	2.58448
Co1-Fe1(x,1/2+y,-1/2+z)	2.58448
Co1-Fe1(x,-1/2+y,1/2+z)	2.58448
Co1-Fe1(x,1/2+y,1/2+z)	2.58448
Co1-Fe1(1/2+x,y,-1/2+z)	2.58448
Co1-Cr1(x,1/2+y,1/2+z)	2.58448
Co1-Cr1(-1/2+x,1/2+y,z)	2.58448
Co1-Cr1(1/2+x,y,-1/2+z)	2.58448
Co1-Cr1(-1/2+x,y,1/2+z)	2.58448
Co1-Cr1(1/2+x,y,1/2+z)	2.58448
Co1-Cr1(1/2+x,-1/2+y,z)	2.58448
Co1-Cr1(1/2+x,1/2+y,z)	2.58448
Co1-Cr1(x,1/2+y,-1/2+z)	2.58448
Co1-Cr1(x,-1/2+y,1/2+z)	2.58448
Fe1-Pd1(-1/2+x,y,1/2+z)	2.58448
Fe1-Pd1(1/2+x,y,1/2+z)	2.58448
Fe1-Pd1(1/2+x,-1/2+y,z)	2.58448
Fe1-Pd1(-1/2+x,1/2+y,z)	2.58448
Fe1-Pd1(1/2+x,1/2+y,z)	2.58448
Fe1-Pd1(x,1/2+y,-1/2+z)	2.58448
Fe1-Pd1(x,-1/2+y,1/2+z)	2.58448

Distance

Bond Def.	Distance, Å
Fe1-Pd1(x,1/2+y,1/2+z)	2.58448
Fe1-Pd1(1/2+x,y,-1/2+z)	2.58448
Fe1-Ni1(-1/2+x,1/2+y,z)	2.58448
Fe1-Ni1(x,1/2+y,1/2+z)	2.58448
Fe1-Ni1(x,-1/2+y,1/2+z)	2.58448
Fe1-Ni1(x,1/2+y,-1/2+z)	2.58448
Fe1-Ni1(1/2+x,1/2+y,z)	2.58448
Fe1-Ni1(1/2+x,-1/2+y,z)	2.58448
Fe1-Ni1(1/2+x,y,1/2+z)	2.58448
Fe1-Ni1(-1/2+x,y,1/2+z)	2.58448
Fe1-Ni1(1/2+x,y,-1/2+z)	2.58448
Fe1-Fe1(-1/2+x,1/2+y,z)	2.58448
Fe1-Fe1(1/2+x,y,1/2+z)	2.58448
Fe1-Fe1(1/2+x,-1/2+y,z)	2.58448
Fe1-Fe1(x,-1/2+y,1/2+z)	2.58448
Fe1-Fe1(x,1/2+y,-1/2+z)	2.58448
Fe1-Fe1(1/2+x,1/2+y,z)	2.58448
Fe1-Fe1(-1/2+x,y,1/2+z)	2.58448
Fe1-Fe1(1/2+x,y,-1/2+z)	2.58448
Fe1-Fe1(x,1/2+y,1/2+z)	2.58448
Cr1-Pd1(x,-1/2+y,1/2+z)	2.58448
Cr1-Pd1(1/2+x,y,1/2+z)	2.58448
Cr1-Pd1(-1/2+x,y,1/2+z)	2.58448
Cr1-Pd1(1/2+x,y,-1/2+z)	2.58448
Cr1-Pd1(x,1/2+y,1/2+z)	2.58448
Cr1-Pd1(x,1/2+y,-1/2+z)	2.58448

Distance

Bond Def.	Distance, Å
Cr1-Pd1(1/2+x,1/2+y,z)	2.58448
Cr1-Pd1(-1/2+x,1/2+y,z)	2.58448
Cr1-Pd1(1/2+x,-1/2+y,z)	2.58448
Cr1-Ni1(1/2+x,y,1/2+z)	2.58448
Cr1-Ni1(-1/2+x,y,1/2+z)	2.58448
Cr1-Ni1(1/2+x,y,-1/2+z)	2.58448
Cr1-Ni1(1/2+x,-1/2+y,z)	2.58448
Cr1-Ni1(x,1/2+y,1/2+z)	2.58448
Cr1-Ni1(x,-1/2+y,1/2+z)	2.58448
Cr1-Ni1(x,1/2+y,-1/2+z)	2.58448
Cr1-Ni1(-1/2+x,1/2+y,z)	2.58448
Cr1-Ni1(1/2+x,1/2+y,z)	2.58448
Cr1-Fe1(x,-1/2+y,1/2+z)	2.58448
Cr1-Fe1(x,1/2+y,-1/2+z)	2.58448
Cr1-Fe1(1/2+x,1/2+y,z)	2.58448
Cr1-Fe1(-1/2+x,1/2+y,z)	2.58448
Cr1-Fe1(1/2+x,-1/2+y,z)	2.58448
Cr1-Fe1(1/2+x,y,1/2+z)	2.58448
Cr1-Fe1(x,1/2+y,1/2+z)	2.58448
Cr1-Fe1(1/2+x,y,-1/2+z)	2.58448
Cr1-Fe1(-1/2+x,y,1/2+z)	2.58448
Cr1-Cr1(1/2+x,y,-1/2+z)	2.58448
Cr1-Cr1(-1/2+x,y,1/2+z)	2.58448
Cr1-Cr1(1/2+x,y,1/2+z)	2.58448
Cr1-Cr1(1/2+x,-1/2+y,z)	2.58448
Cr1-Cr1(1/2+x,1/2+y,z)	2.58448

Distance

Bond Def.	Distance, Å
Cr1-Cr1(x,1/2+y,-1/2+z)	2.58448
Cr1-Cr1(x,-1/2+y,1/2+z)	2.58448
Cr1-Cr1(x,1/2+y,1/2+z)	2.58448
Cr1-Cr1(-1/2+x,1/2+y,z)	2.58448